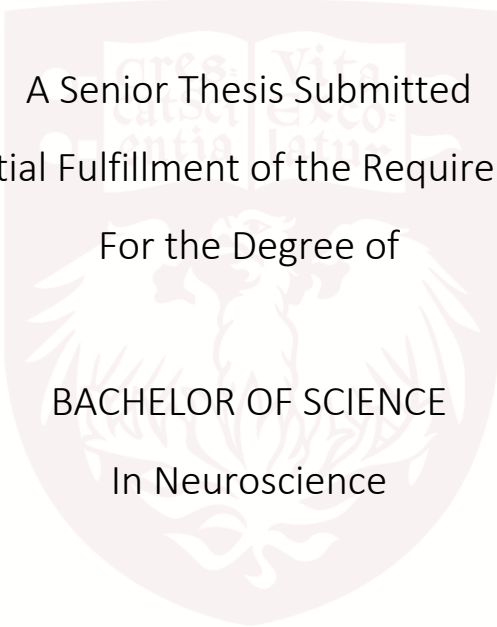


Uncovering Human Proprioceptive Function and Malfunction

Yuri Vieira Sugano



A Senior Thesis Submitted
In Partial Fulfillment of the Requirements
For the Degree of
BACHELOR OF SCIENCE
In Neuroscience

The University of Chicago

2022

Significance statement

Large fiber proprioceptive peripheral afferents are particularly sensitive to demyelinating and inflammatory conditions. Thus, assessment of proprioception is an invaluable tool in diagnostics and disease management involving peripheral neuropathies. So far, psychometric techniques that assess proprioceptive function provide limited information as results tend to be task-specific and nongeneralizable. Physiological techniques are either invasive or expensive. Here I describe a non-invasive task that can assess general proprioceptive function. The task consists of pointing at a target with eyes open and closed. While simple, the task uncovers metrics that highlight normal proprioceptive function and allow determination of malfunction. I conclude that the task is a powerful way to assess proprioceptive function, which in turn serves as a proxy for large fiber health.

Abstract

Currently, standard protocols to quantitatively address proprioception do not exist. Here, I describe a non-invasive task that can provide information about proprioceptive function. A subject points towards a fixed target on a screen and is instructed to maintain the arm position for 60 seconds, with eyes opened or closed. With eyes open, the task is accomplished through the maintenance of shoulder joint position in addition to visual feedback. With eyes closed, the task is under control of shoulder joint position alone. The comparison between a subject's performance with eyes closed and open can be used to assess proprioceptive function. I have established the distribution of metrics such as total excursion, final position, major axis of movement and major oscillatory frequencies in subjects without a known neurological condition to determine a physiological range of proprioceptive function. Subsequently, I tested subjects with known neurological conditions. The metrics reveal differences between trials with and without vision, indicating that they quantify proprioceptive performance. Furthermore, the metrics reveal that the performance of individuals with known neurological conditions falls outside of the physiological range. Thus, we find that the task can reveal proprioceptive malfunction. The task provides an overall evaluation of proprioceptive function as it integrates different proprioceptive domains such as joint position detection, movement detection, and degree of muscle contraction. Therefore, I propose that the new task provides a non-invasive,

inexpensive way to assess proprioceptive function and thus to indirectly assess large afferent fiber health.

Introduction

Broadly, proprioception refers to one's sense of body position. Proprioception is indispensable for movement due to its role in body alignment and fast adjustments in response to outside perturbations. Proprioception was a term coined by Charles S. Sherrington to refer to the senses coming from within body tissues (Sherrington, 1907). Sherrington demonstrated that information about the length of muscles was encoded by the nervous system (Sherrington, 1913). Sherrington's experiments predicted the existence of sensory cells specialized in computing the length of a muscle. These cells have been characterized as proprioceptors, and the sensory endings of these cells are in muscle fibers and in tendons (Tuthill and Azim, 2018). Muscles consist of parallel running fibers. Some of these fibers are intrafusal fibers, sitting between the larger, force-producing, extrafusal fibers. Intrafusal fibers contain the sensory organs of muscles: the muscle spindles. Muscle spindles contain two main types of sensory neurons: type Ia afferents carry information about rates of change, while type II afferents carry information about muscle length (Proske and Gandevia, 2012). In addition, the Golgi tendon organs are sensory organs located in tendons. In Golgi tendon organs, type Ib fibers carry information about muscle tension (Lundberg et al., 1978; Proske and Gandevia, 2012).

Proprioceptive perception is currently evaluated by active reproduction of joint angles or as detection of passive movement (Nagai et al., 2016). Proprioception perception is highly dependent on the task, on the joint/muscle being observed, and on whether the dominant or non-dominant side is being tested (Jong et al., 2005). Current established tasks have important shortcomings, for instance active joint detection requires kinesthetic memory of preestablished positions (Hillier et al., 2015). In addition, the task cannot isolate proprioceptive input from other somatosensory inputs. Passive motion detection tasks suffer from different detection thresholds depending on the joint that is being studied. Even for a given joint, thresholds depend on movement speed. Most importantly, all these tasks only evaluate a subset of proprioceptive function.

Yet, some tasks have been replicated and evaluated in a clinical setting. Of these, the Rivermead Assessment of Somatosensory Perception (RASP) has been demonstrated to be

reliable and provide accurate evaluations (Hillier et al., 2015). When RASP is used to address proprioception, a two-choice discrimination task is used: subjects are asked to report if the joint has been flexed or extended (Winward 2002). However, this dichotomous task is not informative in assessing proprioception alone since movement detection can be intact in individuals with severe proprioceptive loss. Alternative tasks include the modified Thomas splint with a Pearson knee-flexion apparatus, which tests for subject's detection of passive knee joint flexion (Barrett et al., 1991). However, the test is hard to adapt to other joints, and most importantly requires the subject to report the estimated angle, and so it assumes that subjects can estimate and recreate angles visually. As such, while many tasks have been developed to address proprioceptive function, and while some of them have been evaluated in a clinical setting, conclusions are highly dependent on the task and muscle group being studied and results are rarely generalizable (Han et al., 2016). For a given subject, there is no correlation across different tests of either passive movement or joint position detection, and thus current tests do not describe proprioceptive function globally (Grob et al., 2002).

Here, I describe and evaluate a task that can assess proprioceptive function globally. The task depends on all sub-modalities of proprioception: joint angle detection, movement detection, and muscle tension. The task more closely resembles an ecological situation in which proprioceptive feedback is actively engaged, such as holding an object steady in the dark. The task does not rely on passive motion, thus feedback from other somatosensory modalities such as touch or pressure is minimal. I have established physiological ranges for the performance of control subjects without a known neurological condition, and further investigated how select individuals with known neurological disorders perform.

Materials and Methods

A total of 57 subjects without a known medical condition were recruited. Subjects were recruited at the University of Chicago medical campus, and no specific selection criteria were employed. Another 9 subjects with known medical conditions were recruited, either through direct recommendation of their attending physicians, or through the control recruiting process, but were later determined to have an underlying medical condition. All subjects were informed about the task and the goal of the study. Subjects did not receive compensation for participating.

The study was approved by the University of Chicago Institutional Review Board (IRB17-1058).

Experimental Design

The task was performed with the aid of a device developed in our laboratory. The device consists of a target with concentric rings for orientation. A camera was placed behind the target at a fixed distance. Subjects were asked to wear a white glove and the tip of the finger was marked with a colored-marker for color-assisted detection. Subjects were asked to sit comfortably on a chair, then the height was adjusted so that the shoulder was approximately at the same height of the target. The subject was asked to point straight ahead, and the target was placed to be as close to the fingertip as possible, without contact.

The initial trials were randomized (2x2 conditions left versus right hand, eyes open or closed). The subsequent trials were performed in a specific order relative to the first: same hand, different visual condition; opposite hand, same visual condition; then opposite hand, opposite visual condition. The subject was first asked to point to the target, then asked to close their eyes in trials without vision. The start of the recording marks the beginning of the trial. Recordings continued for 60 seconds. Then, the subject was told to rest the arm for another 60 seconds. Subsequent trials were performed in a similar manner.

Videos were recorded at 25 frames per second. Subsequently, the computer vision software Ethovision v. 9.0 was used to obtain position coordinates. Coordinates were subdivided into horizontal and vertical components. The following metrics were calculated from these coordinates:

The position at the first frame was established as the starting position. The distance between the final and the initial position was the *final position*. The *average horizontal position* $\bar{x}$ and the *average vertical position* $\bar{y}$ were calculated. Similarly, the *average distance to the center* across the trial was calculated. As a measure of spread, the *horizontal variance* and the *vertical variance* were calculated. In addition, the *horizontal range* is the distance between the most extreme points in the y-axis, similarly the *vertical range* is the distance between the most extreme points in the x-axis. The velocity was estimated as the discrete time derivative of the position. The average of the velocity was calculated in the horizontal (*horizontal velocity*) and vertical (*vertical velocity*) components as well as the absolute magnitude (*average velocity*). The percentage of time in each quartile was determined. Lastly, a directional, standard deviational

ellipse was calculated. In short, this was an ellipse that covered 99.9% of the data points. The ellipse's centroid was centered at the center of mass of the distribution, $(\bar{x}, \bar{y})$ and therefore was also a metric for the two-dimensional average position in the task. The tangent of the angle of rotation was calculated as the inverse tangent of the two-dimensional standard deviation $\tan^{-1}(\sigma_{1,2})$ calculated using the mean-squared-error of the horizontal ($\tilde{x}^2$) and vertical ($\tilde{y}^2$) distributions.

$$\sigma_{1,2} = \left(\frac{\left(\sum_{i=1}^n \tilde{x}_i^2 + \sum_{i=1}^n \tilde{y}_i^2 \right) \pm \sqrt{\left(\sum_{i=1}^n \tilde{x}_i^2 - \sum_{i=1}^n \tilde{y}_i^2 \right)^2 + 4 \left(\sum_{i=1}^n \tilde{x}_i \tilde{y}_i \right)^2}}{2n} \right)^{\frac{1}{2}}$$

The tangent of the angle of rotation of the ellipse represents the dominant axis of movement in each trial. Other angular metrics were calculated: *average position angle* and *final position angle*, where 0 degrees corresponds to the +x axis, -90 degrees corresponds to the -y axis and 90 degrees corresponds to the +y axis.

Statistical Analyses

The distribution of the metrics was often skewed, particularly for trials with eyes open. A Shapiro-Wilk test was used to investigate whether each metric was obtained from a non-normal distribution. For cases in which that was true, a Wilcoxon signed rank was used to determine whether two distributions came from the same underlying distributions (for instance between trials with and without vision). Effect sizes were reported according to Tomczak & Tomczak (2014).

Each subject with a known condition was a unique data point, and no grouping of these subjects was made, despite some having received the same diagnostic. Thus, each subject with a known condition was compared to the distribution that we established for control subjects. All data manipulation and statistical analyses were done in R version 4.1.2.

Since many variables were calculated for each subject, a principal component analysis (PCA) was performed as a first step. Using metrics from control subjects only, a principal component analysis was performed using the *prcomp* function in R, after data was normalized. Then, the linear transformation corresponding to the first principal component (PC1) was applied to the results of subjects with a known neurological condition to obtain a PC1 score for these subjects. This was done to not introduce potential outliers in the PCA estimates. Lastly, a fast-

Fourier transform was performed according to Cooley and Tukey (1965) to obtain dominant frequencies in horizontal and vertical movement components. Raw data, scripts used in data manipulation and in statistical analyses are openly available by request.

Results

A total of 57 subjects had not been diagnosed with a neurological condition. The results from these 57 subjects were grouped together as a control group. Of these, 30/57 (53%) subjects identified as female, whereas 27/57 (47%) identified as male. The average age was 32 (standard deviation = 18) years. A breakdown of demographics is shown in Table I.

A principal component analysis was performed on all the metrics extracted from the raw data. The first principal component (PC1) accounts for 35% of the variability in the data. The correlation between PC1 and any of the individual variables was never greater than 0.22, indicating that no single variable can account for a significant portion of the variability in proprioceptive performance. PC1 was used in subsequent analysis along with the most relevant metrics revealed by the principal component analysis.

Table I: Demographics of subjects without a known neurological condition (controls)

Gender	
Male	27 (47%)
Female	30 (53%)
Age	
18-19	15 (26%)
20-29	22 (39%)
30-39	5 (8.8%)
40-49	2 (3.5%)
50-59	4 (7.0%)
60+	8 (14%)
No response	1 (1.7%)

Establishing a physiological range for control subject

A sample from one of the controls, representative of what a control trial typically looked like, is shown in Figure 1. Subjects tended to perform well with eyes open, rarely deviating from the center. The median horizontal range was 0.9 (IQR 0.7-1.2) and the median vertical range was 0.97 (IQR 0.7-1.4). In contrast, with eyes closed, these medians were 2.3 (1.6-3.8) and 3.0 (2.0-4.8), respectively. An ellipse

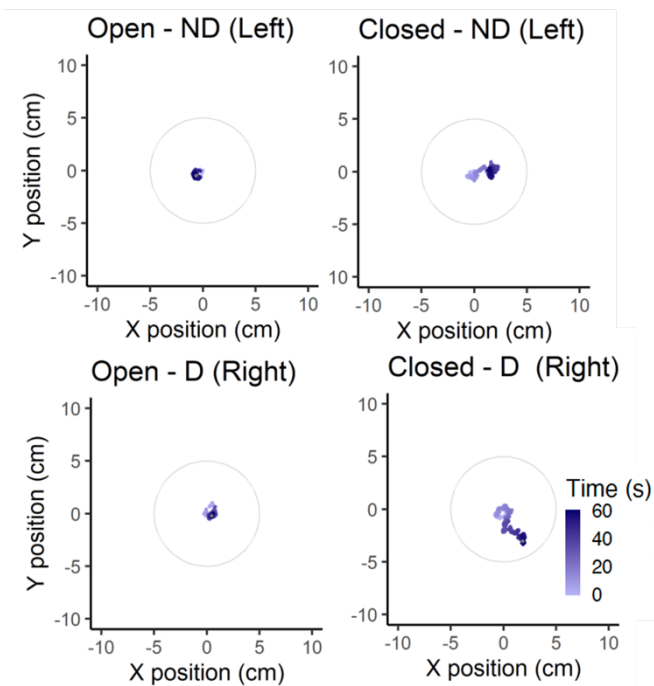


Figure 1: A typical control performance. The four conditions consisted of two trials with eyes open (right) and two trials with eyes closed (left), and two trials with the non-dominant (ND) hand (top, left hand in this case) and two trials with the dominant (D) hand (bottom, right hand in this case). Trials lasted for 60 seconds. Time indicated in blue gradient where white is the start and dark blue is the end of the trial.

encompassing 99.9% of the data points is another useful metric. The median area of the ellipse was 2.9 (1.6-5.3) cm² with eyes open versus 25.5 (12.7-47.8) cm² with eyes closed. A linear regression showed that only eyes open versus closed can account for differences in all the metrics. Hand side, dominance, sex, or age did not account for differences in any of the metrics.

The first principal component (PC1) differed between trials with or without vision (Figure 2A; Wilcoxon signed rank: $V(N = 57) = 5519$, $P < 0.001$, $r^2 = 0.62$). The final position in trials without vision was greater than in trials with vision (Figure 2B; $V(N = 57) = 866$, $P < 0.001$, $r^2 = 0.63$). Similarly, the average distance to the center was greater without vision (Figure 2C; $V(N = 57) = 865$, $P < 0.001$, $r^2 = 0.62$). Other metrics were also greater in trials without vision such as the horizontal range, vertical range, ellipse centroid and ellipse area. Interestingly, while the total excursion was greater without vision, the magnitude and the spread of the difference was the smallest among all metrics (Figure 2D; $V(N = 57) = 1917$, $P = 0.0002$, $r^2 = 0.35$). Through visual inspection, the distribution of total excursion is the same between trials with eyes open or closed. However, the paired-samples statistics indicate that total excursion tended to decrease for each given subject when eyes were closed. Yet, the graph highlights that the

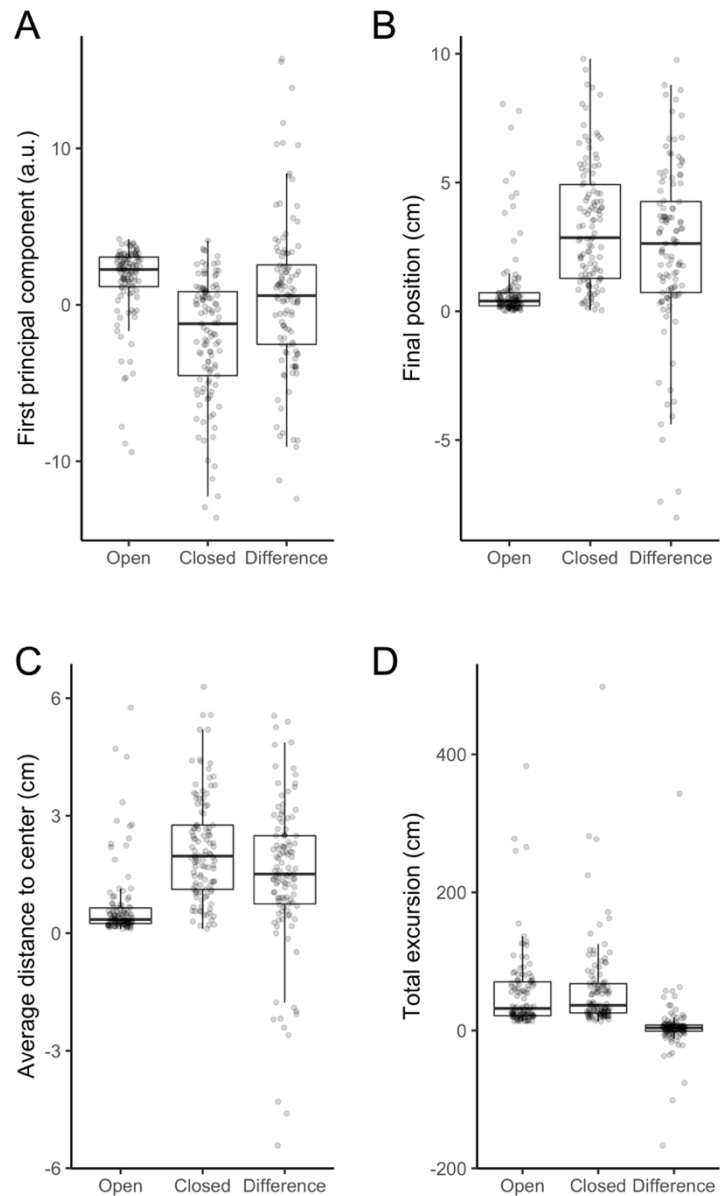


Figure 2: Determination of control performance range. Boxplots comparing trials with eyes open and closed and their difference. (A) First principal component in arbitrary units. (B) Final position calculated as the distance from the final position to the initial position. (C) Average distance to center across entire trial and (D) Total excursion as the integration of all distance vectors. (B-D) in units of centimeter. Boxplots show median (middle line), 25th and 75th percentiles (bottom and top of the boxes, respectively). Whisker (horizontal line) extends to last data point inside a distance of 1.5xIQR (75th – 25th percentile) above and below the box. Raw data shown as gray dots in the background.

magnitude of the decrease is small. Total excursion was also not correlated with PC1 ($r^2 = 0.04$). As such, the total distance traveled was not greatly affected by a lack of vision.

Next, I investigated the angular pattern of the movement of control subjects. There was no difference in the angle at the final position between trials with eyes open or closed (Figure 2a; $V(N = 57) = 3427$, $P = 0.55$, $r^2 = 0.06$). In figure 3A, the dashed blue line represents a final angle of -90° , a final position that would be expected if gravity was acting alone. While most subjects did not move straight down, there was a trend towards the lower half circle, particularly in trials with eyes closed (Figure 3B).

The rotation of the ellipse encompassing 99.9% of the data points represents the angle of the axis where most of the movement occurred. The distribution of the major axis angle was bimodal in both conditions, with peaks around 65° and -65° (Figure 3C). The troughs at 0° and $90^\circ/-90^\circ$ in the figure suggest that a small number of trials consisted of purely horizontal (0°) or

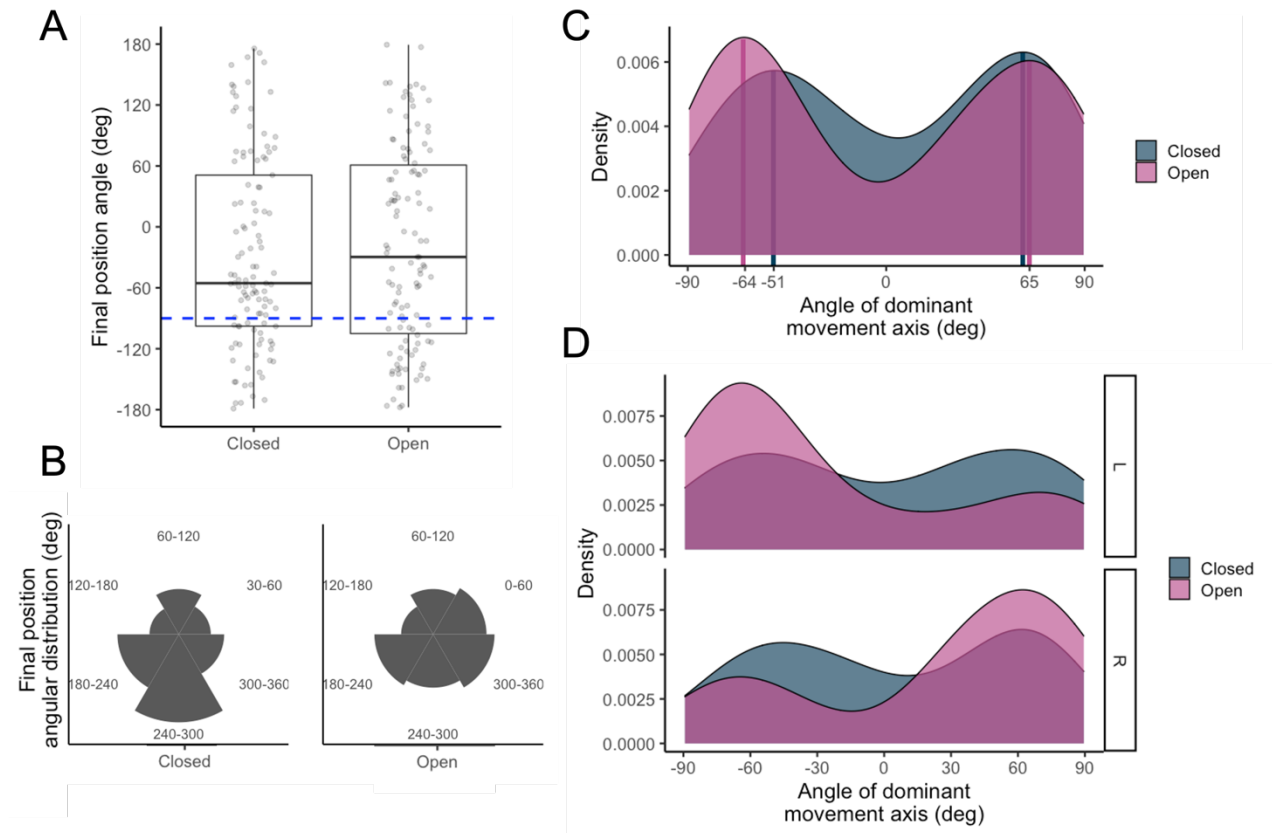


Figure 3: Determination of angular distribution range for controls. (A) comparison between the final position angle in trials without vision (eyes closed) and with vision (eyes open). Boxplot follows the style of Figure 1. Dashed blue line marks the -90° angle. (B) Angular distribution of the final position. A downward skew (towards the negative y-axis) can be observed in trials without vision. (C) Angle corresponding to the major axis of movement in each trial. A bimodal distribution is observed in both conditions. (D) Same as (C) but separated by left (L) and right (R) hands.

vertical ($90^\circ/-90^\circ$) movements. Furthermore, it seems like in trials without vision there is a slight shift of the dominant axis toward the negative. Since negative and positive directions can represent medial or lateral movements depending on the side of the hand being used, it is likely that the bimodality of the distributions is due to opposing hands. Figure 3D shows the distribution for each hand separately. The distribution for either hand in trials without vision is largely the same (Figure 3D, dark blue). In contrast, the distribution in trials with vision was dependent on the hand. When the left hand was used, the movement was mostly in the negative angular direction. When the right hand was used, the movement was mirrored to the positive angular direction.

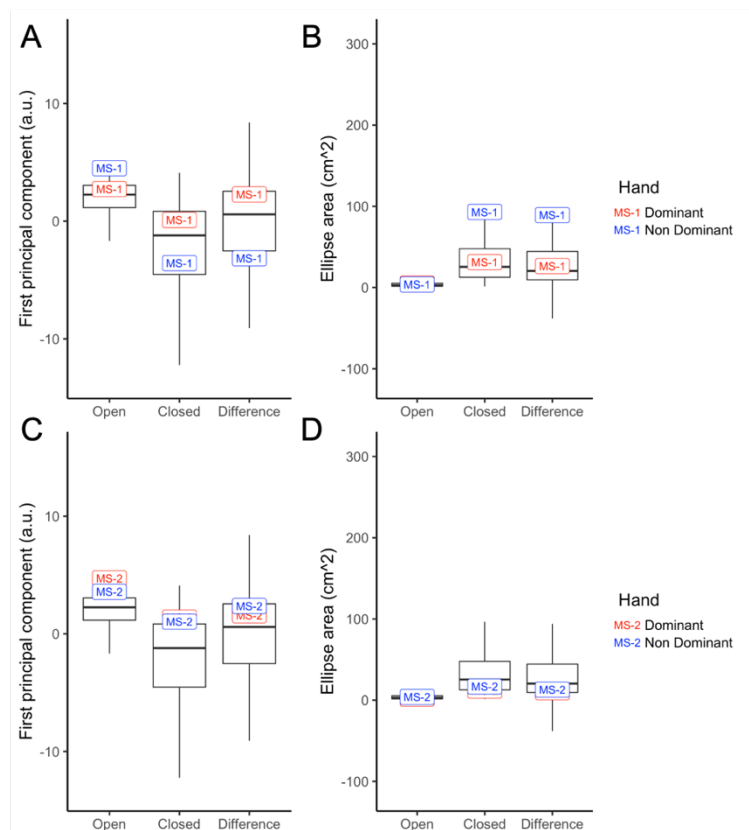
Comparing the performance of subjects with a neurological condition to that of controls

A total of 9 subjects had been identified with a neurological condition. 4/9 of the subjects with known neurological condition had been diagnosed with multiple sclerosis (MS). Other diagnoses were muscle weakness (MW), Tourette's syndrome (TS), Ehlers-Danlos syndrome (ED), chronic inflammatory demyelinating polyneuropathy (CIDP) and a rare, atypical form of hereditary sensory autonomic neuropathy (subject KS). 1/8 of the subjects were male (second subject with multiple sclerosis, MS-2). Average age was 43 years (standard deviation = 14, range 22-64).

Multiple sclerosis

The first of the MS subjects had PC1 results within the control range (Figure 4A). However, the area of the directional ellipse fell at the limit of controls when the subject had their eyes closed, and only for the non-dominant hand (Figure 4B; area = 93 cm^2 , closer to 91 percentile of controls), suggesting that under these conditions the

Figure 4: Not all subjects with multiple sclerosis (MS) had abnormal results. A-D boxplots follow similar format of Figure 2 and show the distribution for controls. (A) First principal component (PC1) of the first MS subject, MS-1. PC1 was within the control range. (B) Area of the ellipse encompassing 99.9% of the data. MS-1 fell at the higher limit of the control range with eyes closed, only with the non-dominant hand. (C & D) Similar to A & B, but for subject 2 (MS-2). MS-2's performance, including the area of the ellipse, was comparable to that of controls.



subject covered a larger area. No other metrics were abnormal; thus, the subject finished the trial within control range (Figure 5A). The second MS subject had both PC1 and ellipse metrics within control range (Figure 4C-D). The third and fourth subjects also had these metrics within control range, indicating that the performance for subjects with MS is not homogeneous.

As additional evidence, I compared the final position for all four MS subjects (Figure 5). While all fell within control range, some MS subjects clearly demonstrated a separation in performance between the non-dominant and dominant hand with eyes closed. Subject 2 (MS-2, Figure 5B) had the best performance, with both hands falling right on the median of controls. A gradual increase in difference can be seen for subject 4 (MS-4, Figure 5D), subject 1 (MS-1, Figure 5A) and subject 3 (MS-3, Figure 5C). Interestingly, MS-3 seemed to perform slightly better than controls with the dominant hand. All the differences were observed in trials with eyes closed and in the difference between eyes open and closed, indicating that the changes were due to proprioceptive function and not motor function.

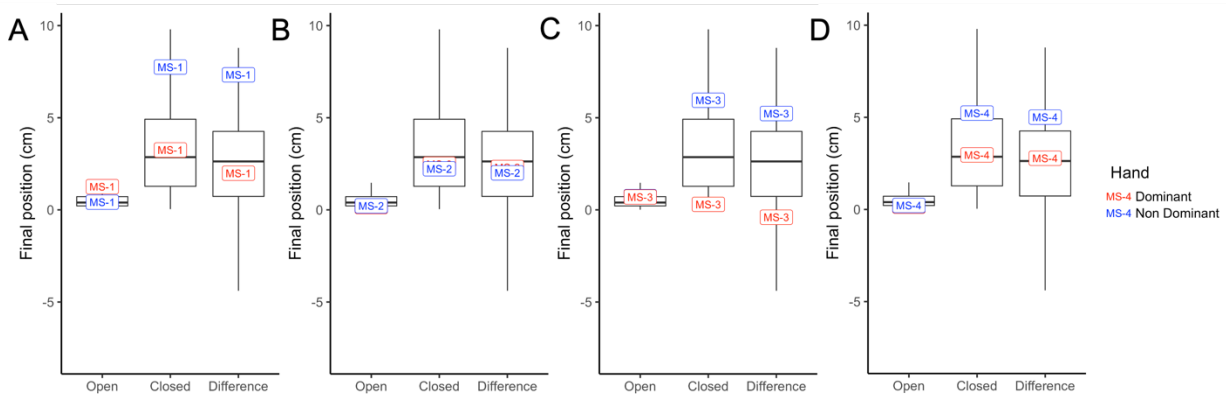


Figure 5: The performance of subjects with multiple sclerosis (MS) is heterogeneous. All subjects were compared in a single metric, final position. A-D boxplots follow similar format of Figure 2 and show the distribution for controls. (A) MS-1 had worse performance with the non-dominant hand in the trial without vision. (B) MS-2 had results close to the control median. (C) MS-3 had slightly worse performance with the non-dominant hand but performed better than 75% of controls with the dominant hand. (D) MS-4 had only a slightly worse performance with the non-dominant hand, with the dominant hand falling right at the control median.

Muscle weakness

One of the subjects had been diagnosed with muscle weakness. The overall performance (PC1) as well as the final position were not different from controls (data not shown). However, in trials with eyes closed, the range of the vertical movement was larger than controls with the non-dominant hand (Figure 6A). In contrast, the range of the horizontal movement was like that of

controls (Figure 6B). The centroid of the ellipse encompassing 99.9% of the data was farther away from the center than the normal range for controls (Figure 6C; distance to center = 5.1 cm, control 95th percentile = 4.3 cm). However, the area of the ellipse was not different (area = 43 cm², control IQR = 12.7-47.8). As such, it appears that the abnormalities are due to an unusual downward displacement. No other metrics showed any difference in this subject.

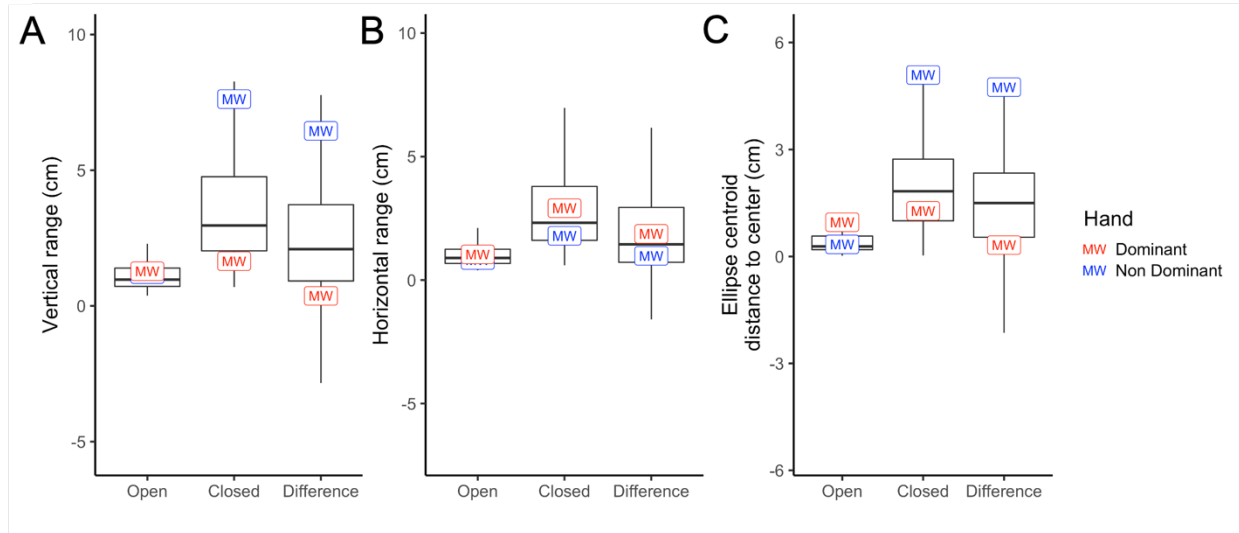


Figure 6: The vertical range was abnormal for an individual with muscle weakness (MW). A-C boxplots follow similar format of Figure 2 and show the distribution for controls. (A) The vertical range of movement is different between the dominant and the non-dominant hand, in trials with eyes closed. While the weakness supposedly affects both hands, the performance with the dominant hand was better than 75% of controls. (B) The horizontal range, in contrast, was comparable to that of controls. (C) The centroid of the ellipse encompassing 99.9% of the data points was farther away from the center than that of controls, in trials with eyes closed and with the non-dominant hand.

Tourette's syndrome

The subject who had been diagnosed with Tourette's syndrome demonstrated an unusual profile. First, there were abnormalities in total excursion, a metric that rarely differed between controls, even between trials with eyes open and closed (Figure 6A; eyes open: subject total excursion = 296 cm (dominant hand); 356 cm (non-dominant hand); control 97.5th percentile = 261; eyes closed: subject = 216 cm (dominant hand); 198 cm (non-dominant hand); control 97.5th percentile = 235 cm). The area of the ellipse (Figure 6B) and the range of vertical motion (Figure 6C) were all abnormal when compared to controls, but only for trials with eyes closed and with the non-dominant hand.

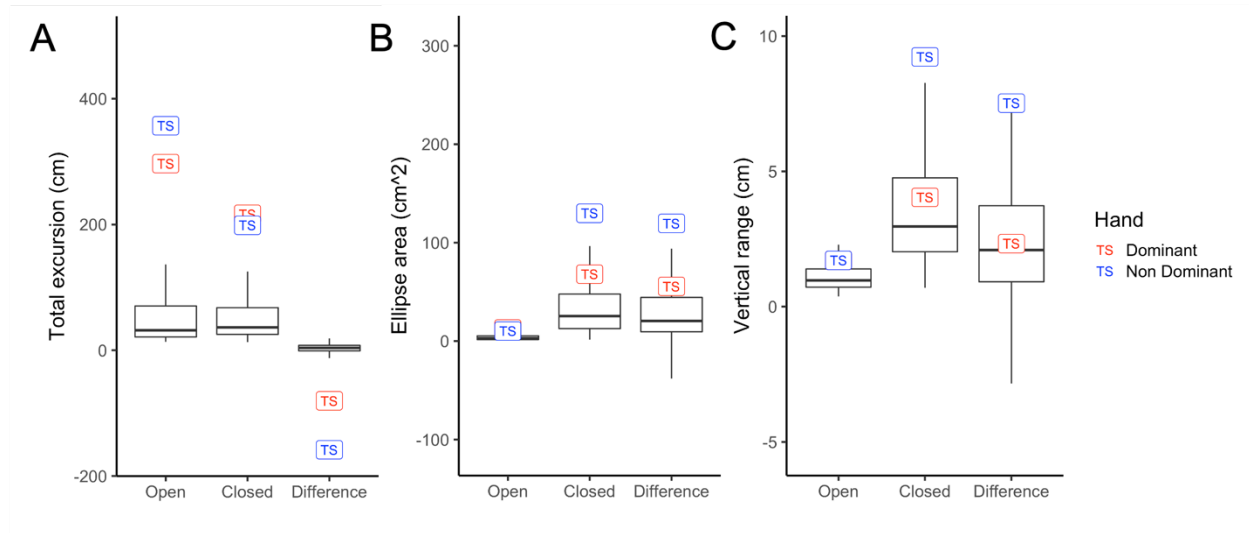


Figure 7: Tourette's syndrome is associated with abnormal total excursion, regardless of vision. A-C boxplots follow similar format of Figure 2 and show the distribution for controls. (A) Total excursion was higher for TS, particularly with eyes open. (B) Ellipse area was greater than that of controls in trials without vision and with the non-dominant hand. (C) Vertical range was higher than that of controls in trials without vision and with the non-dominant hand.

Ehlers-Danlos and chronic inflammatory demyelinating polyneuropathy

Like most of the prior results, the abnormalities observed in subjects diagnosed with Ehlers-Danlos (ED) and chronic inflammatory demyelinating polyneuropathy (CIDP) are related to deviation in performance between the dominant and the non-dominant hand in trials with eyes closed.

For the subject diagnosed with ED, the vertical range of motion was greater than that of controls for trials using the non-dominant hand (Figure 8A; vertical range = 7.3 cm; control IQR = 2.0-4.8 cm; control 92nd percentile = 7.1 cm). Interestingly, the horizontal range of motion was greater than that of controls for trials using the dominant hand (Figure 8B; horizontal range = 6.9 cm; control IQR = 1.6-3.8 cm; control 96th percentile = 6.8 cm).

For the subject diagnosed with CIDP, the differences were observed in trials with the non-dominant hand. The area of the ellipse was greater than that of controls (ellipse area = 116 cm²; control IQR = 13-48 cm², control 94th percentile = 115 cm²). The centroid of the ellipse was farther away to the center than the centroid of controls (centroid distance = 5.0 cm; control IQR = 1-2.7 cm; control 96th percentile = 4.7 cm).

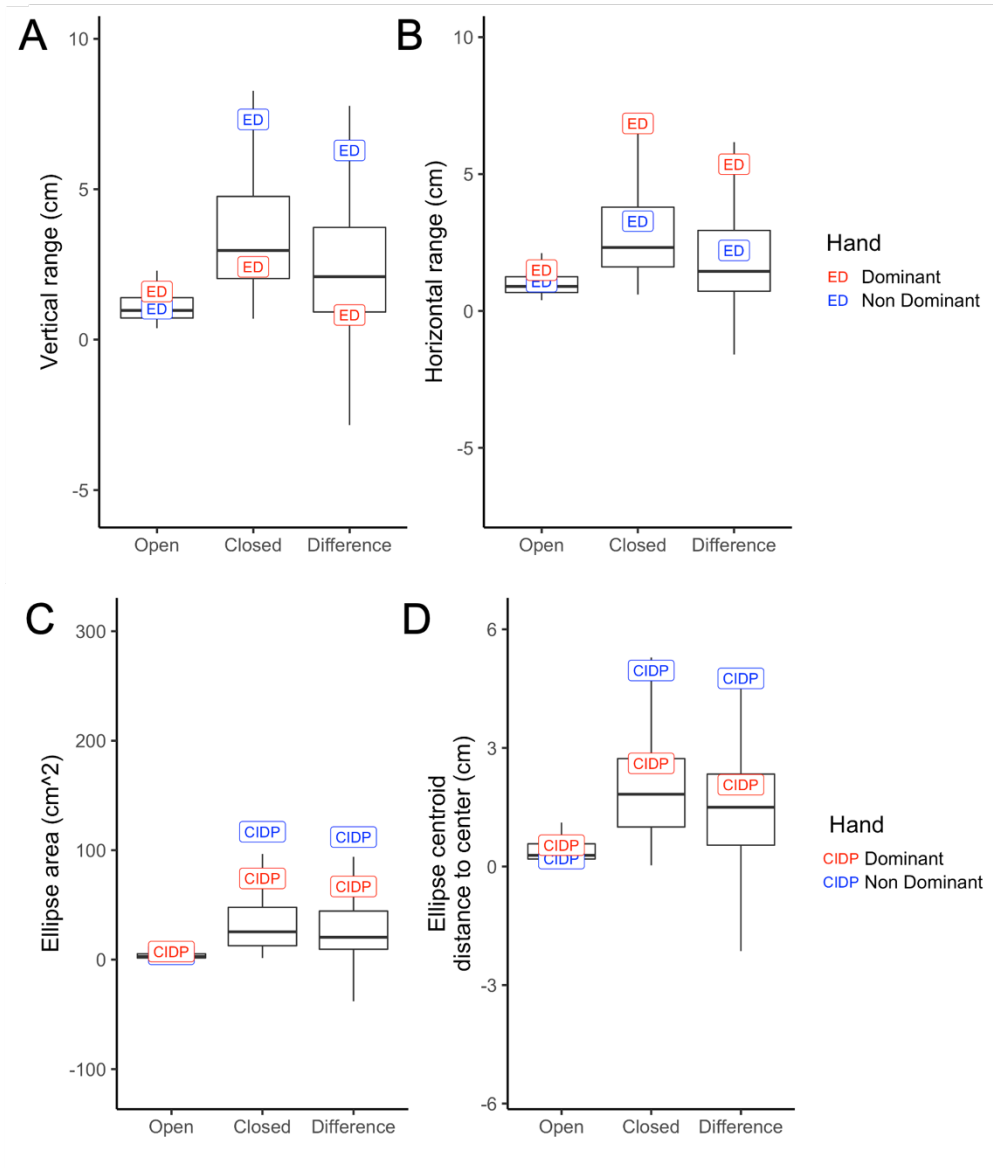


Figure 8: Ehlers-Danlos and chronic inflammatory demyelinating polyneuropathy are associated with divergent performance between dominant and non-dominant hands. A-D boxplots follow similar format of Figure 2 and show the distribution for controls. (A) Vertical range is higher for ED in trials with eyes closed, with the non-dominant hand. (B) Horizontal range, however, is higher in trials with the dominant hand. (C) The area of the ellipse is greater than that of controls, for CIDP subjects in trials with eyes closed and with the non-dominant hand. (D) The distance of the centroid to the center is also greater in subjects with CIDP.

A rare form of hereditary sensory autonomic neuropathy

KS is a subject who has been identified to have a severe form of hereditary sensory autonomic neuropathy that completely affects all somatosensory fibers, including large diameter fibers associated with proprioception. KS was the only subject, including all other subjects with neurological condition, to obtain abnormal metrics in PC1, indicating that the overall

performance in the task was different from that of controls (Figure 9A; eyes closed: KS's dominant hand = -11.2, KS's dominant hand = -12.6, control IQR = -4.5-0.8; control 2.5th percentile = -11.3). As expected from PC1, KS's performance was visibly different to that of controls in all metrics (Figure 9B-F). In all metrics, KS is far above the 97.5th percentile for controls.

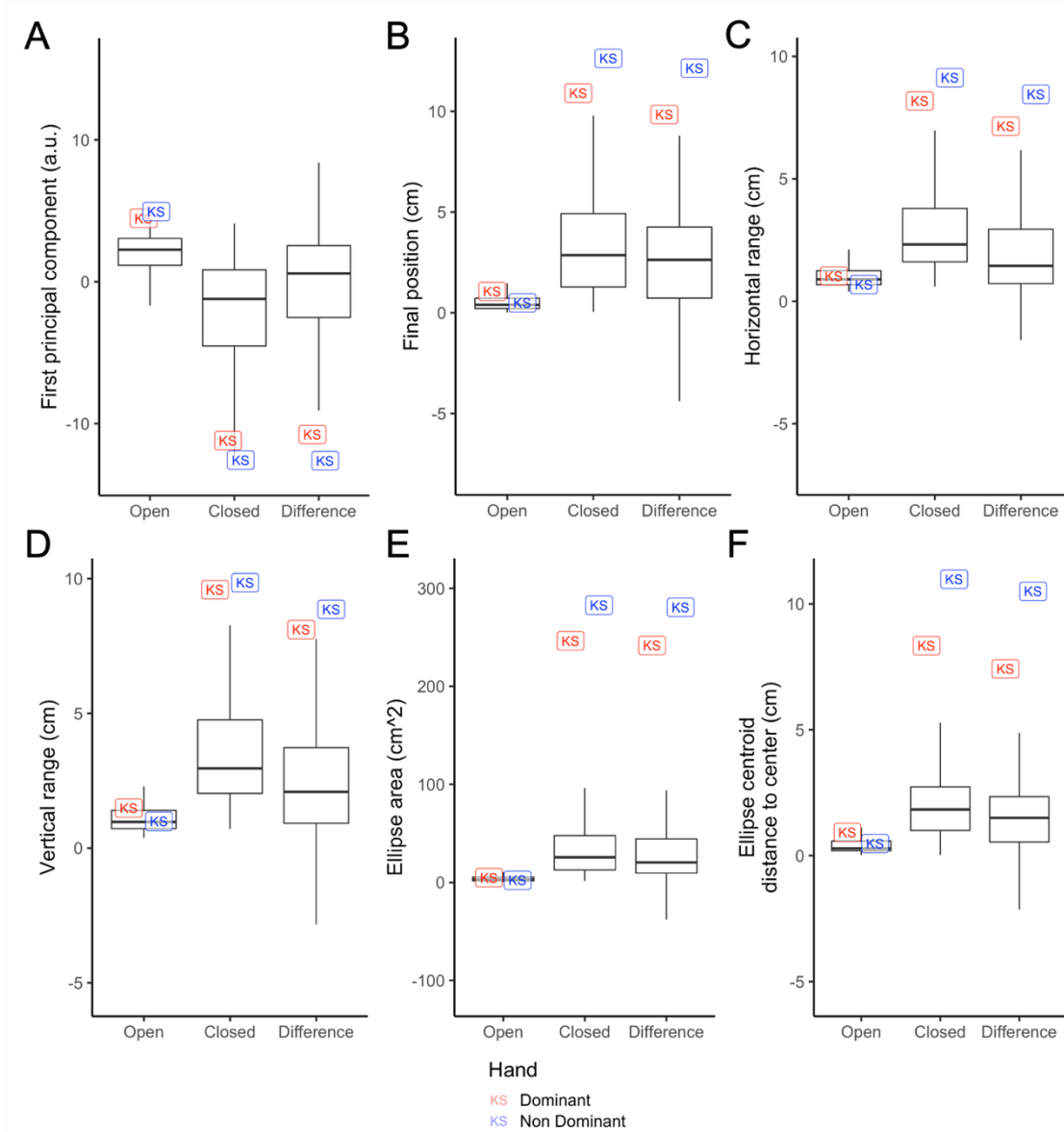


Figure 9: For a subject with complete lack of proprioception, all the metrics are abnormal. A-D boxplots follow similar format of Figure 2 and show the distribution for controls. (A) PC1 was below the 2.5th percentile for controls in trials with eyes closed, with both dominant and non-dominant hands. Since PC1 is comprised of a linear combination of all the metrics, this suggests that the entire performance was different between KS and controls. (B) Final position was greater for KS than for controls. Both horizontal (C) and vertical (D) range were also higher for KS than for controls. Lastly, both the ellipse area (E) and the distance of the center of the ellipse (F) were greater for KS.

Dominant frequencies in movement: a possible detection for tremor

As a last goal, I established the range of dominant frequencies in the horizontal and vertical components for controls using a fast-Fourier transform of the x-position across time and y-position across time, respectively. The dominant frequency for all controls was 0 Hz, consistent with a complete absence of oscillatory behavior (Figure 10). The analysis up to the 30th dominant frequency did not demonstrate any frequencies above 3 Hz, and rarely demonstrated frequencies above 2 Hz. A large portion of dominant frequencies are observed around 0.25 Hz.

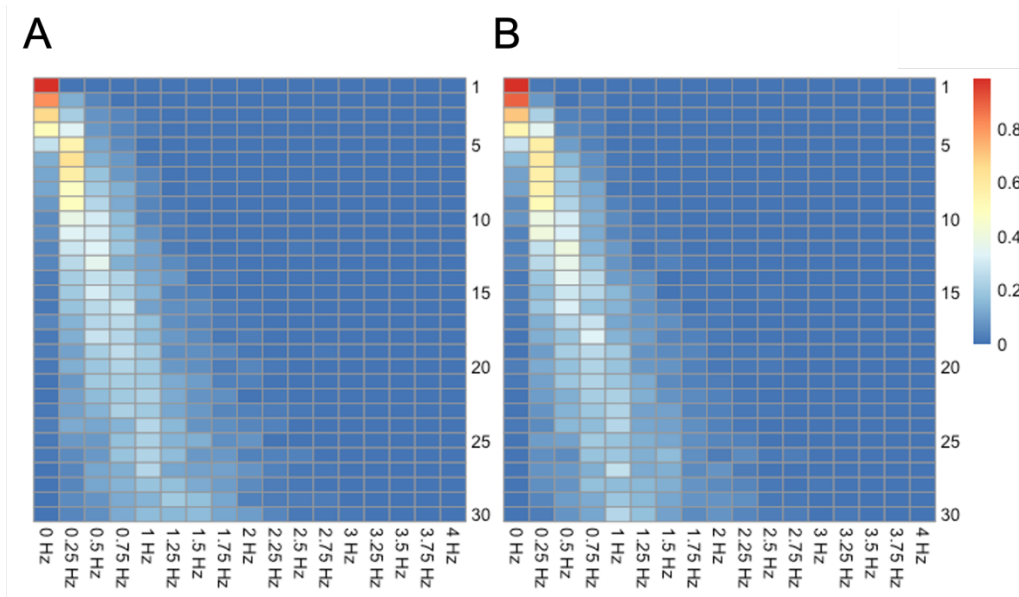


Figure 10: Subjects without a known neurological condition do not have dominant oscillatory movements. In both panels, each column corresponds to a 0.25 Hz frequency bin. Rows represent the first 30 dominant frequencies. In each row, frequencies were normalized so that all columns in a row add up to one. Each n row represents the distribution of frequencies for the n -th dominant frequency. Each row can be seen as a distribution histogram where the density is given by the color. Red is highest density (1), and blue is the lowest (0). (A) Dominant frequencies in the horizontal direction (x-axis). The first dominant frequency is 0 Hz for all subjects, corresponding to a non-oscillatory movement. Up to the 30th dominant frequency, no frequencies larger than 3 Hz are observed. (B) Same as A, with similar results. No conspicuous differences between horizontal and vertical distributions can be seen.

All subjects with known neurological conditions also had the first dominant frequency at 0 Hz, suggestive that no major tremor was observed. Yet, some of the subjects differed from the control distribution shown above. MS-4, one of the subjects with multiple sclerosis, had a second dominant frequency at 0.4 Hz in a trial with eyes open and MS-3 had a third dominant frequency at 0.8 Hz with eyes closed. Subsequent dominant frequencies for MS-3 were around 0.8 (0.72-0.88), suggesting that there was likely an underlying oscillatory movement at 0.8 Hz (1.25 seconds).

MS-1 had a fifth dominant frequency at 1 Hz. None of the subjects reported tremor as one of their symptoms.

Discussion

Here, I established a physiological range in several metrics in for control subjects. The first principal component (PC1) is the best estimator for a subject's overall performance in the task. The other metrics are not perfectly correlated and each of the metrics likely represents different aspects of proprioceptive function. Then, I compared the performance of subjects with several known neurological conditions to the performance of controls. All neurological conditions had at least one result that fell at the limit or outside of the control range. Interestingly, different conditions displayed abnormalities in different metrics. This agrees with the hypothesis that different metrics highlight different aspects of proprioceptive function.

Although control subjects were able to maintain the position accurately with eyes open, the finger in trials with vision was not stationary. In trials with eyes open, the fine tuning of the movement is guided by the visual cueing of the tip of the finger pointing straight to the target. However, the total excursion was nearly the same in trials with and without vision. It appears the visual, somatosensory, and motor systems are providing constant feedback and minor corrections are constantly being made. The observation of a dominant frequency at 0.25 Hz suggests that this correction likely happens in the order of 4 second cycles.

Subjects consistently performed worse without vision. The just noticeable difference for shoulder joint angle discrimination in a passive movement task is between one and two degrees (Elangovan et al., 2014). Yet, the observed range of movement was greater than this. Assuming an arm length of 60 cm, two degrees correspond to 2 cm. Most subjects had ranges greater than this in trials with eyes closed. It is likely that the continuous nature of the task and the absence of an external force (which also activates touch and pressure receptors) increase the just noticeable difference considerably. Proprioception is a sensory modality that is comprised of several components. Within muscle fibers. Type II afferents carry information about muscle length; thus, type II afferents are engaged in a task such as pointing to a target and maintaining the position. Type Ia afferents respond optimally to rates of change and thus detect dynamic muscle stretch. In trials with eyes closed, type I and type II afferents work together to maintain the arm extended at the initial shoulder angle. The results indicate that proprioception alone is not sufficient to

maintain position within two to three centimeters, whereas vision allows the position to be maintained under one centimeter.

The task also identified a natural, preferred orientation of movement for the shoulder at approximately 65°. This seemed to be true for both hands and for trials with and without vision. Interestingly, in trials with vision alone, a side preference was observed. Most individuals using the right hand moved at a 65° major axis, whereas individuals using the left hand moved at a -65° major axis. This suggests a mirrored preferred movement in which the arm goes medial as it goes down, and laterally as it goes up. The orientation was apparently random in trials with eyes closed.

Taken together, the task reveals several independent metrics that may highlight proprioceptive function. When compared to the distribution of controls in these metrics, the task can also signal to the presence of malfunction. Interestingly, different conditions had different presentations, and the same condition (MS) also had varying degrees of the same presentation. Thus, the task can likely differentiate between different aspects of proprioceptive and motor function, as well as to the degree of health of the fibers underlying these processes.

Most abnormalities consisted of divergent behavior between the non-dominant and dominant hands, particularly in trials without vision. This implies that even subjects with abnormal results, still were able to perform well in the task. Thus, it appears that the task has a low detection threshold and that one does not need to have significant damage to large fibers to show abnormal results.

The results indicate that the task may show abnormalities in proprioceptive function in multiple sclerosis, a disease affecting the central nervous system. Indeed, the task does not measure peripheral fiber health alone, but rather the entire somatosensory-motor apparatus involved in pointing at a target and maintaining the position. Proprioception is known to be affected in multiple sclerosis, yet most of the literature focus on balance and posture (Prosperini et al., 2010; Yang & Liu, 2020). This is because difficulties with posture and balance have far greater consequences than issues with distal joint angle detection. However, it is also likely that issues with posture and balance come at later stages of the disease. Thus, the task described here may provide a way to evaluate proprioceptive function before it becomes a life-threatening complication. It appears that the results were indeed associated with proprioception rather than the motor system as the abnormalities were hand-specific and limited to trials with eyes closed.

Unfortunately, I did not have any background information for any of the MS subjects regarding the relative stage of the MS, whether treatment was effective or if there were any other sensory or motor symptoms that had been described.

The results also highlight a potential for diagnostic. While the sample sizes are small, for controls but particularly for subjects with a known neurological condition, the variability in the metrics is striking. The potential for diagnostic will be evaluated through testing more subjects with known neurological conditions, but also by gathering more information on the current state of their condition. If the observations can be replied (for instance, the vertical range being abnormal for the dominant hand and the horizontal range being abnormal for the non-dominant hand), then these results can be factored into a decision-tree like algorithm to assist physicians with diagnostics.

References

- Barrett D, Cobb A, Bentley G (1991) Joint proprioception in normal, osteoarthritic and replaced knees. *The Journal of Bone and Joint Surgery British* volume 73-B:53–56.
- Cooley JW, Tukey JW (1965) An algorithm for the machine calculation of complex Fourier series. *Math Comp* 19:297–301.
- Grob KR, Kuster MS, Higgins SA, Lloyd DG, Yata H (2002) Lack of correlation between different measurements of proprioception in the knee. *The Journal of Bone and Joint Surgery British* volume 84-B:614–618.
- Han J, Waddington G, Adams R, Anson J, Liu Y (2016) Assessing proprioception: A critical review of methods. *Journal of Sport and Health Science* 5:80–90.
- Hillier S, Immink M, Thewlis D (2015) Assessing Proprioception: A Systematic Review of Possibilities. *Neurorehabil Neural Repair* 29:933–949.
- Jong A de, Kilbreath SL, Refshauge KM, Adams R (2005) Performance in Different Proprioceptive Tests Does Not Correlate in Ankles With Recurrent Sprain. *Archives of Physical Medicine and Rehabilitation* 86:2101–2105.
- Lundberg A, Malmgren K, Schomburg ED (1978) Role of joint afferents in motor control exemplified by effects on reflex pathways from Ib afferents. *The Journal of Physiology* 284:327–343.

- Nagai, T., Allison, K. F., Schmitz, J. L., Sell, T. C., & Lephart, S. M. (2016). Conscious proprioception assessments in sports medicine: how individuals perform each submodality. *Sports Med: SM Online Scientific Resources*, 1-13.
- Proske U, Gandevia SC (2012) The Proprioceptive Senses: Their Roles in Signaling Body Shape, Body Position and Movement, and Muscle Force. *Physiological Reviews* 92:1651–1697.
- Prosperini L, Leonardi L, De Carli P, Mannocchi ML, Pozzilli C (2010) Visuo-proprioceptive training reduces risk of falls in patients with multiple sclerosis. *Mult Scler* 16:491–499.
- Sherrington CS (1907) ON THE PROPRIO-CEPTIVE SYSTEM, ESPECIALLY IN ITS REFLEX ASPECT. *Brain* 29:467–482.
- Sherrington CS (1913) REFLEX INHIBITION AS A FACTOR IN THE CO-ORDINATION OF MOVEMENTS AND POSTURES. *Exp Physiol* 6:251–310.
- Tomczak M, Tomczak E (2014) The need to report effect size estimates revisited. An overview of some recommended measures of effect size. 1:8.
- Tuthill JC, Azim E (2018) Proprioception. *Current Biology* 28:R194–R203.
- Winward CE, Halligan PW, Wade DT (2002) The Rivermead Assessment of Somatosensory Performance (RASP): standardization and reliability data. *Clin Rehabil* 16:523–533.
- Yang F, Liu X (2020) Relative importance of vision and proprioception in maintaining standing balance in people with multiple sclerosis. *Multiple Sclerosis and Related Disorders* 39:101901.